

1. Maps and their structure

A short course in the relationships behind the results.

Definition 3.1. Vector spaces

A vector space is a set closed under addition and scalar multiplication.

The argument follows from Theorem 3.4.

Definition 3.2. Linear maps

A map T is linear when $T(au + bv) = aT(u) + bT(v)$.

The argument follows from Theorem 3.5.

Definition 3.3. Kernel and image

The kernel is the set of vectors sent to zero. The image is the range of T .

The argument follows from Theorem 3.6.

Theorem 3.4. Rank-Nullity

(i) The kernel and image of A are subspaces. (ii) If V is finite dimensional, $\dim \ker A + \dim \operatorname{im} A = \dim V$.

The argument follows from Theorem 3.7.

2. Bases and dimension

A short course in the relationships behind the results.

Corollary 3.5. Injectivity criterion

A linear map is injective if and only if its kernel contains only zero.

The argument follows from Theorem 3.8.

Lemma 3.6. Basis extension

Every linearly independent list extends to a basis of a finite dimensional space.

The argument follows from Theorem 3.9.

Definition 3.7. Direct sums

If U and W intersect only at zero, their sum is a direct sum.

The argument follows from Theorem 3.10.

Definition 3.8. Isomorphisms

An invertible linear map identifies two vector spaces of equal dimension.

The argument follows from Theorem 3.11.

3. The quotient construction

A short course in the relationships behind the results.

Proposition 3.9. Composition

The composition of two linear maps is linear.

The argument follows from Theorem 3.12.

Theorem 3.10. Dimension formula

The dimension of $U + W$ equals $\dim U + \dim W$ minus $\dim$ of their intersection.

The argument follows from Theorem 3.1.

Definition 3.11. Quotient spaces

The quotient V/U consists of cosets of U , with the induced vector operations.

The argument follows from Theorem 3.2.

Theorem 3.12. First isomorphism theorem

The quotient of V by $\ker T$ is isomorphic to the image of T .

The argument follows from Theorem 3.3.

4. A proof of Rank-Nullity

The dimensions that disappear, and the dimensions that remain.

Let $T: V \rightarrow W$ be linear, with V finite dimensional.

Choose a basis $v(1), \dots, v(k)$ of the kernel of T .

By Basis extension, complete this to a basis $v(1), \dots, v(n)$ of V .

We claim $T(v(k+1)), \dots, T(v(n))$ is a basis of the image of T .

Linearity shows that these vectors span the image.

To prove independence, suppose a linear combination of them is zero.

Then the corresponding combination of $v(k+1), \dots, v(n)$ lies in $\ker T$.

By Kernel and image, this is a combination of $v(1), \dots, v(k)$.

Independence of the full basis forces every coefficient to vanish.

Thus $\dim \ker T = k$, while $\dim \operatorname{im} T = n - k$.

Adding the two dimensions gives $\dim V = n$, as required.